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RESEARCH ARTICLE

Zero-Shot Voice Cloning Text-to-Speech for Dysphonia Disorder Speakers

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ABSTRACT Deep neural network based text-to-speech (TTS) technology has brought advances in speech synthesis approaching the quality of human speech sounds. Zero-shot voice cloning TTS is a system that accepts input in the form of text and a few seconds of a sample of the target speaker's voice to produce speech sound waves similar to the target speaker's voice. Some of the latest zero-shot voice cloning TTS studies still focuses on normal human voices. However, this technology still has limitations for individuals with speech disorders such as dysphonia. We observe that our baseline zero-shot TTS model applied to the dysphonia domain still has poor performance on the following aspects: speaker similarity, intelligibility or clarity of speech, and speech sound quality. This research develops 24 zero-shot voice cloning TTS models to observe which models can improve the baseline model performance on the dysphonia domain. We propose four categories to change the baseline model architecture and setting: input-level text sequences (grapheme, phoneme, or combination of grapheme-phoneme), speaker embedding type (speaker encoder or speaker model), speaker embedding position (at TTS encoder of at TTS encoder and decoder), and loss function (without or with speaker consistency loss). The experimental results show that the best model is the one uses the following configuration settings: a combination of grapheme-phoneme-level text sequences, speaker model as the speaker embedding, placing the speaker embedding at the TTS encoder only, and adding speaker consistency loss to the frame-level speech loss. Compared to the baseline model, our proposed best model is able to improve speaker cosine similarity (COS), speech intelligibility (CER), and speech sound quality (MOS) performance in the domain of dysphonia speech disorders by 0.197, 0.55%, and 0.244, respectively. When compared with the original voice of dysphonia disorder speakers, the best model also increases the speech intelligibility and quality of the speech sounds by 13.45% and 0.22, respectively.

INDEX TERMS Deep neural network, dysphonia, speaker consistency loss, speaker encoder, speaker model, text-to-speech, voice cloning, zero-shot learning.

I. INTRODUCTION

Communication is one of the basic human activities and voice is the most natural verbal communication that is easily understood by normal humans [1]. Spoken communication has advantages over written communication because the speaker's emotions and expressions can be conveyed better. However, it becomes an obstacle for humans with speech disorders such as dysphonia. Dysphonia speech disorder is characterized by a laryngospasm that can lead to breathy and/or creaky speech sound that is unclear and more difficult to understand [2]. Such a speech disorder significantly affects a person's social life, emotional well-being, and career.

In the Society 5.0 era, the development of artificial intelligence and massive changes in the digital world in the industrial revolution 4.0 are aimed at providing comfort in human life [3]. One of these is the development of human computer interaction (HCI) which requires the ability of a computer to process speech sound signals so that it can communicate and interact with humans or as a device that helps normal humans to communicate with people with visual, hearing, and speech impairments. Speech

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signal processing covers several fields, including: automatic speech recognition, automatic speaker identification, automatic speaker verification, text-to-speech, speech translation, spoken language understanding, speech representation, dialogue system, and speech enhancement [4]. Unfortunately in a series of surveys, individuals with dysphonia speech disorder have described their difficulty in using these voice-based technologies because these systems rarely understand their speech [2].

Text-to-speech (TTS) studies how computers can read text or symbols and speak them aloud. This study covers three things, namely: the reading process, the speaking process, and various issues encountered when the computer performs this task [5]. Several deep neural network (DNN)-based TTS studies have achieved excellent quality in recent years [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. Not only those single-speaker TTS systems aforementioned, the multi-speaker TTS systems also show satisfactory results such as [18], [19], [20], [21], [22], [23], [24], [25], and [26]. However, those TTS technologies are generally tested for normal speakers. Individuals with speech disorders such as dysphonia cannot use the TTS system directly without further testing.

To produce good speech sound synthesis, the TTS system requires a huge training data containing clean, high-quality recorded speech sounds to train the model. Therefore, it becomes problematic to develop a TTS system for people with speech disorder such as dysphonia. It cannot use the dysphonic speech voice recordings data as the references for good quality speech during the model training. One approach that can be used is zero-shot speaker adaptation TTS [27], [28], [29], [30], [31]. This approach is able to synthesize speech from target speakers without requiring the target speakers' speech recordings to be available in the training data. The zero-shot speaker adaptation TTS utilizes pre-trained speaker encoders that have been trained separately for speaker recognition tasks, such as automatic speaker identification and automatic speaker verification (ASV). Furthermore, the pre-trained speaker encoder generates speaker embedding that can be used by the TTS model to adapt to the new speaker through a transfer learning process. All mentioned zero-shot speaker adaptation TTS models are reported for normal human voice cloning.

In this paper we present research on zero-shot voice cloning TTS tested on the dysphonia speech disorder domain. For the base TTS model we use Tacotron-2 architecture [12]. To handle zero-shot voice cloning, we adopt a DNN-based d-vector speaker encoder using a generalized-end-to-end (GE2E) model for speaker verification [32]. For the baseline model we use the TTS model with the addition of a d-vector speaker encoder to generate target speaker representation as done in [27]. This baseline model is reported to have a poor performance for dysphonia disorder domain in the following aspects: speaker similarity, intelligibility or clarity of speech, and speech sound quality.

To improve the performance of the zero-shot voice cloning TTS system on the target dysphonic speakers, we propose four categories changes applied to the baseline model: input-level setting of the text sequence, speaker embedding type, speaker embedding position, and loss function. Our baseline model uses grapheme-level text sequences as input of the model, speaker embedding generated from target speaker speech examples by a pre-trained speaker encoder network, speaker embedding position at the TTS encoder, and a standard loss function as used in Tacotron-2. For the input-level category of text sequences we add new options, phoneme-level and a combination of graphemes-phoneme with a probability of 0.5. We assume that the new input-level options using phoneme or a combination of phoneme-grapheme can increase the synthesized speech intelligibility. For the speaker embedding type category used during model training, we add a new option to use a speaker model for each target speaker. The speaker model is calculated from the average speaker representation of all target speaker utterances available in the training data. The new option is expected to increase the speaker similarity of the synthesized speech. Next, for the speaker embedding position category, we add a new option to place the speaker embedding at the TTS encoder and decoder at the same time. This option is also expected to increase the speaker similarity. Finally, for the loss function category we add a new option to incorporate speaker consistency loss with the speech-frame loss assuming this can also increase the speaker similarity. In total, we develop 24 different zero-shot voice cloning TTS models based on the settings of the aforementioned categories.

We apply two learning stages to train our models: 1) speaker encoder network training for speaker verification and 2) TTS model training for voice cloning speech synthesis. The training process involves two domain datasets: the source domain dataset containing the recordings of normal human speech voices that are used to train the model and the target domain dataset containing the recordings of normal human and dysphonic speech voices that are used for model testing. In Stage 1, the speaker encoder network is trained using the source domain dataset and it is used directly without fine-tuning to produce a speaker representation of the target speakers. In Stage 2, we use transfer learning of the pre-trained speaker encoder without fine-tuning during the training of the zero-shot voice cloning TTS models on the source domain dataset. The transfer learning without fine-tuning of the speaker encoder allows the TTS models to synthesize speech sounds from the target domain dataset using a few seconds of sound samples from the target normal and dysphonic speakers.

Our experiments use the English LibriTTS-Train360 dataset [33] as the source domain to train the speaker encoder for the speaker verification task and to train the TTS models for the voice cloning speech synthesis task. As the target domain, we use English UncommonVoice dataset [2] which contains voice recordings of normal and dysphonic

speakers. From the ablation study of the whole experiments carried out in this research, we observe several findings. First, the use of phoneme-level and a combination of grapheme-phoneme-level text sequences is able to increase the speech intelligibility if accompanied by the application of the speaker consistency loss. Second, using speaker model calculated for each target speaker as the speaker embedding can improve the speaker similarity if the position of the speaker embedding remains at the TTS encoder only. Third, adding speaker embedding information to the TTS encoder and decoder is able to increase the speaker similarity if we use speaker embedding generated by the speaker encoder from a speech sample of the target speaker. Fourth, adding the speaker consistency loss to the frame-level speech loss can improve the speaker similarity in all cases. Fifth, the contribution of each category to the improvement in speech sound quality is not quite clear because it depends on the combination of some or all category settings. Finally, of the 24 models developed, the best model is the one that applies the following settings: a combination of grapheme-phoneme-level text sequence, speaker model as the speaker embedding, placing speaker embedding information at the TTS encoder only, and adding speaker consistency loss into the frame-level speech loss.

Overall, the contribution of this research can be summarized as follows:

1. Presenting a zero-shot voice cloning TTS study on the target domain dataset containing the speech recordings of the target normal and dysphonic speakers. This study differs from the most TTS studies which focus on normal human voice cloning.
2. Proposing new modifications applied to the baseline TTS model grouped in four categories setting options: text sequence input-level setting, speaker embedding type, speaker embedding position, and loss function. This study aims to find the best model among the different 24 zero-shot voice cloning TTS models developed. The ablation study to know the contribution of each category setting to the model performance is also presented.

The remainder of this paper is organized as follows. Section II reviews works related to this research. Section III presents the proposed zero-shot voice cloning TTS architecture and its training stages. Section IV provides the implementation details of our TTS models. Section V presents the experimental results and analyses, and Section VI concludes the research.

II. RELATED WORKS

To enhance the speech quality of speech disorder speakers, there are two categories of speech reconstruction approaches: voice conversion (VC) and text-to-speech (TTS) [34]. The former is to adjust the speech signals produced by the speech disorder speakers to be more natural and intelligible, such as recent works on dysarthria disorder [34], [35]. The latter is to build personalized TTS systems by cloning the voice of the pre-recorded patients' speech, which is the focus of our work. If the patients' speech recordings before their speech deteriorates are available, the TTS model can be trained using these data. Otherwise, if there are only deteriorated speech recordings available then another training method needs to be applied for training the TTS model. Using such speech data for model training produces poor quality synthesized speech. Our work applies zero-shot voice cloning method to build personalized TTS system for dysphonic speakers.

There are several previous studies on zero-shot voice cloning TTS that use speaker embedding generated by the pre-trained speaker encoder without fine-tuning the TTS model [27], [28], [30], [31], [36]. These studies are tested on the normal human speech dataset. Different from the previous studies, our research aims to apply the TTS model to clone the voice of dysphonic speakers as in recent study [37] and test our model to the dysphonia speech dataset. However, unlike [37] that focuses on the impact of limited phonetic content applied on a speaker encoder used by a TTS system, our study focuses on the voice cloning TTS system to produce synthesized speech with more clarity and intelligibility without sacrificing the speaker similarity. Moreover, [37] presented a preliminary work on the speaker encoder experiments using normal speaker dataset and did not yet apply on the dysphonic speaker dataset. They also did not provide the impact to the whole TTS model.

Some zero-shot voice cloning studies [27], [28], [36], [37] use x-vector speaker encoder [38], [39], [40], [41] and another study [31] uses H/ASP speaker encoder [42], while our approach adopts a DNN-based d-vector speaker encoder using the GE2E model [32] as also used by [30]. This is because DNN-based d-vector speaker encoder is reported to work well on low-resource domain [43] such as dysphonic speakers domain used in our research.

As reported by [27], [28], and [36] the performance of their zero-shot voice cloning TTS models are still unsatisfactory: the synthetic speech produced still has incorrect accents and different prosodic characteristics from the target speaker speech. We adopt zero-shot voice cloning TTS [27] combined by GE2E speaker encoder [32] as our baseline model. To improve the TTS model performance we proposed new modifications in four categories: input-level text sequence, speaker embedding type, speaker embedding position, and loss function. The proposed new approach presented in this paper is an extended study of our previous work [30] on low-resource language domain.

III. METHODS

In this section, we present the proposed zero-shot voice cloning TTS architecture, training strategy, and four setting categories applied to our baseline model. The baseline model adopts Tacotron-2 [12], GE2E speaker encoder [32], and WaveGlow vocoder [44]. As for the training strategy, we apply transfer learning and zero-shot learning in two stage training. We propose four setting categories in our model: input-level of the text sequence, speaker embedding type, speaker embedding position, and loss function. Each category

has two or three setting options. In total, we develop 24 different TTS models based in these settings, one of which is the baseline model.

A. MODEL ARCHITECTURE

Fig 1 shows the proposed Tacotron-2 based zero-shot voice cloning TTS model architecture that is also used during the model inference. The TTS model consists of mel-spectrogram prediction network and a vocoder. The mel-spectrogram prediction network is an attention-based encoder decoder that predicts mel-spectrogram $Y' = (\boldsymbol{y}'_1, \ldots, \boldsymbol{y}'_{TY})$ given textual input sequences $X = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{Tx})$ and a speech sample mel-spectrogram $M = (\boldsymbol{m}_1, \ldots, \boldsymbol{m}_{TM})$ from the target speaker. The mel-spectrogram prediction networks consist of a text encoder, a speaker encoder, and a mel-spectrogram decoder. Finally, the vocoder generates an output speech waveform $W = (w_1, \ldots, w_{T_W})$ from the predicted mel-spectrogram $Y' = (\boldsymbol{y}'_1, \ldots, \boldsymbol{y}'_{TY})$. Each component of the TTS model architecture is explained below.

1) TEXT ENCODER

The text encoder takes input text sequences $X = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_{Tx})$ and converts them into a hidden representation $H = (\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{Tx})$ as follows:

$$H = (\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{Tx}) = \text{TextEncoder}_{\theta_{\text{te}}}(X), \quad (1)$$

where θ_{te} is the parameters of the text encoder network and T_X is the number of tokens in the normalized text. As shown from the figure, we provide three options to tokenize the text sequences: grapheme-level, phoneme-level, or a combination of grapheme-phoneme-level.

2) SPEAKER ENCODER

The speaker encoder produces a d_S-dimensional d-vector speaker embedding. We separately train the speaker encoder network using GE2E loss for speaker verification task. During the TTS training process we keep the pre-trained speaker encoder's parameter weights without fine-tuning. The speaker encoder generates d-vector speaker embedding $\boldsymbol{q}$ based on the target speaker mel-spectrogram $M = (\boldsymbol{m}_1, \ldots, \boldsymbol{m}_{TM})$ calculated using the following formula:

$$\boldsymbol{q} = \text{SpeakerEncoder}_{\theta_{\text{se}}}(M), \quad (2)$$

where θ_{se} is the speaker encoder network parameters.

During the model inference process there is only one option of speaker embedding, which is produced by speaker encoder network. However, this is not the case during the training process that is explained in Section III-B.

3) MEL-SPECTROGRAM DECODER

The mel-spectrogram decoder consists of attention network, autoregressive (AR) decoder, pre-net, and post-net. In general it produces mel-spectrogram acoustic features by processing all encoder outputs and previous decoder output. Attention network and autoregressive decoder process the hidden representation $H = (\boldsymbol{h}_1, \ldots, \boldsymbol{h}_{Tx})$ output from text encoder concatenated with speaker embedding $\boldsymbol{q}$ to generate predicted mel-spectrogram $Y = (\boldsymbol{y}_1, \ldots, \boldsymbol{y}_{TY})$ and stop token $Z = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_{TY})$.

For each decoder step t, AR decoder calculates the output mel-spectrogram frame $\boldsymbol{y}_t$, the new hidden state decoder $\boldsymbol{s}_t$, and the stop token $\boldsymbol{z}_t$. The decoder stops producing mel-spectrogram frames if it is indicated by the stop token. As already mentioned earlier, we have two options to where we put the speaker embedding: only at the encoder network or at both the encoder and the decoder pre-net. First option, the speaker embedding is not fed to pre-net, therefore the decoder produces the mel-spectrogram frame $\boldsymbol{y}_t$, the new hidden state decoder $\boldsymbol{s}_t$, and the stop token $\boldsymbol{z}_t$ as formulated below:

$$\boldsymbol{s}_t, \boldsymbol{y}_t, \boldsymbol{z}_t = \text{ARDecoder}_{\theta_d}\left(\boldsymbol{s}_{t-1}, \text{PreNet}_{\theta_{\text{pn}}}\left(\boldsymbol{y}_{t-1}\right), \boldsymbol{c}_t\right), \quad (3)$$

while the second option feeds the speaker embedding into pre-net as formulated as follows:

$$\boldsymbol{s}_t, \boldsymbol{y}_t, \boldsymbol{z}_t = \text{ARDecoder}_{\theta_d}\left(\boldsymbol{s}_{t-1}, \text{PreNet}_{\theta_{\text{pn}}}\left(\left[\boldsymbol{y}_{t-1}; \boldsymbol{q}\right]\right), \boldsymbol{c}_t\right), \quad (4)$$

where θ_d is the autoregressive decoder parameters, θ_{pn} is the pre-net parameters, $\boldsymbol{s}_{t-1}$ is the previous state, $\boldsymbol{y}_{t-1}$ is the previous output, $\boldsymbol{c}_t$ is the attention context vector, and $\boldsymbol{q}$ is the speaker embedding. The context vector $\boldsymbol{c}_t$ is calculated to determine the most important encoder step from all encoder outputs, using attention scheme as follows:

$$\boldsymbol{c}_t, \boldsymbol{\alpha}_t = \text{Attention}_{\theta a}\left([H; \boldsymbol{q}], \boldsymbol{s}_{t-1}, \boldsymbol{\alpha}_{t-1}\right), \quad (5)$$

where θ_a is the attention network parameters, $\boldsymbol{\alpha}_t$ is the weight of the attention map. Attention map is a mapping probability between decoder step $t \in T_Y$ with each encoder step $i \in T_X$ generated using location-sensitive attention mechanism. Then, a post-net layer consisting of stacked CNN consumes $Y = (\boldsymbol{y}_1, \ldots, \boldsymbol{y}_{TY})$ to get $Y' = (\boldsymbol{y}'_1, \ldots, \boldsymbol{y}'_{TY})$ by adding the residual prediction as follows:

$$Y' = Y + \text{PostNet}_{\theta_p}(Y), \quad (6)$$

where θ_p is the post-net parameters. This network is used to improve the overall mel-spectrogram reconstruction.

4) VOCODER

This study uses WaveGlow as a vocoder to convert mel-spectrograms $Y' = (\boldsymbol{y}'_1, \ldots, \boldsymbol{y}'_{TY})$ into sound waves $W = (w_1, \ldots, w_{T_W})$ as follows:

$$W = \text{WaveGlow}_{\theta_w}\left(Y'\right), \quad (7)$$

where θ_w is the WaveGlow parameters. WaveGlow is trained separately from mel-spectrogram prediction networks.

B. MODEL TRAINING

While the flow of the TTS model inference is shown in Fig.1, the flow of the model training is shown in Fig.2. We apply two training stages to our zero-shot TTS model: 1) Speaker encoder training for ASV (bottom); 2) Zero-shot TTS training (top). All combinations of options chosen from each model

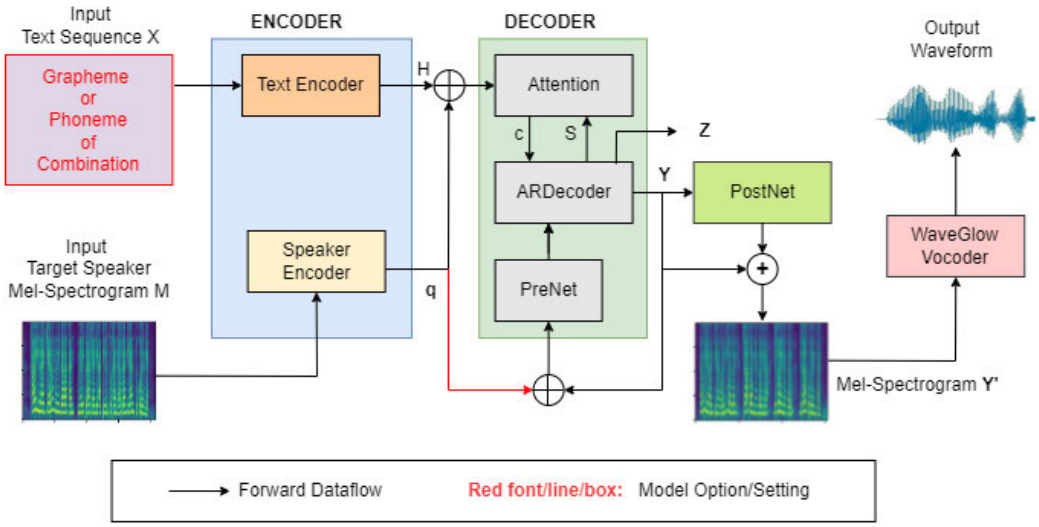


FIGURE 1. Zero-shot voice cloning TTS model inference. Given a text sequence X and the mel-spectrogram of the target speaker's speech sample M, the model predicts the output mel-spectogram Y′ of the text X and generates an output waveform. There are three alternative options for the text sequence: graphme-level, phoneme-level, or a combination of both with probability 0.5. There are two alternative positions of the speaker embedding: at encoder only or at both encoder and decoder.

category are also shown in the figure by red font/line/box. As we already explained previously, there are 24 combinations on how we train our model.

1) SPEAKER ENCODER FOR ASV MODEL TRAINING ON SOURCE DOMAIN

In Stage 1, we train speaker encoder for speaker verification task on the source domain dataset. The speaker encoder model is trained using GE2E loss [32] by making a speaker embedding vector as close as possible to the speaker's centroid and as far as possible from other speakers' centroids, especially the closest one.

A batch of mel-spectrograms created from $N_{\text{spk}} \times N_{\text{utt}}$ utterances of N_{spk} speakers (N_{utt} utterances for each speaker) is used to calculate similarity matrix $S_{ij,k}$, as follows:

$$\boldsymbol{S}_{ij,k} = \begin{cases} \mathrm{f}_{\cos}\left(\boldsymbol{q}_{ij}, \boldsymbol{c}_i^{(-j)}\right) & \text{if } k = i \\ \mathrm{f}_{\cos}\left(\boldsymbol{q}_{ij}, \boldsymbol{c}_k\right) & \text{otherwise,} \end{cases} \tag{8}$$

where $\mathrm{f}_{\cos}$ is the cosine similarity function with learnable weight and bias, $\boldsymbol{q}_{ij}$ is the d-vector embedding of mel-spectrogram M_{ij} extracted from speaker i utterance j, $\boldsymbol{c}_i^{(-j)}$ is the centroid of speaker i without utterance j, and $\boldsymbol{c}_k$ is the centroid of the speaker k, $k \neq i$.

The model is trained using GE2E loss which is the sum of all vector embedding losses against the similarity matrix $\boldsymbol{S}$ using the following formula:

$$\text{GE2Eloss}\left(\boldsymbol{S}\right) = \sum_{i,j} \mathrm{L}\left(\boldsymbol{q}_{ij}\right), \tag{9}$$

where $\mathrm{L}(\boldsymbol{q}_{ij})$ is the loss in the embedding vector $\boldsymbol{q}_{ij}$, $1 \leq i \leq N_{\text{spk}}$ and $1 \leq j \leq N_{\text{utt}}$, which is formulated as follows:

$$\mathrm{L}\left(\boldsymbol{q}_{ij}\right) = -S_{ij,i} + \log \sum_{k=1}^{N_{\text{spk}}} \exp\left(S_{ij,k}\right). \tag{10}$$

This loss function pushes each embedding vector toward its corresponding centroid and away from all other centroids.

2) ZERO-SHOT TTS MODEL TRAINING ON THE SOURCE DOMAIN

In Stage 2, the pre-trained speaker encoder network parameters in Stage 1 are transferred to the target TTS model without fine-tuning when we train the zero-shot voice cloning TTS model for speech synthesis task. In this training process, all network parameters, except the speaker encoder's, are updated using a chosen loss function.

The data flow formula of the TTS model training is quite similar to the model inference explained in Section III-A. Both have three options of the text sequences and two options of the position to put the speaker embedding information. However, there are some differences due to the teacher-forcing procedure during training process which uses the ground-truth mel-spectrogram Y^{GT} to be fed into the decoder network instead of using the predicted mel-spectrogram Y. As a consequence, Eq. (3) changes into:

$$\boldsymbol{s}_t, \boldsymbol{y}_t, z_t = \text{ARDecoder}_{\theta_d}\left(\boldsymbol{s}_{t-1}, \text{PreNet}_{\theta_{\text{pn}}}\left(\left[\boldsymbol{y}_{t-1}^{\text{GT}}\right]\right), \boldsymbol{c}_t\right), \tag{11}$$

while Eq. (4) changes into:

$$\boldsymbol{s}_t, \boldsymbol{y}_t, z_t = \text{ARDecoder}_{\theta_d}\left(\boldsymbol{s}_{t-1}, \text{PreNet}_{\theta_{\text{pn}}}\left(\left[\boldsymbol{y}_{t-1}^{\text{GT}}; \boldsymbol{q}\right]\right), \boldsymbol{c}_t\right). \tag{12}$$

As it is shown in Fig. 2 (top), we have two options of learning objective during the model training: 1) The first option only uses the frame-level reconstruction loss function and 2) The second option integrates frame-level reconstruction loss and utterance-level speaker consistency loss.

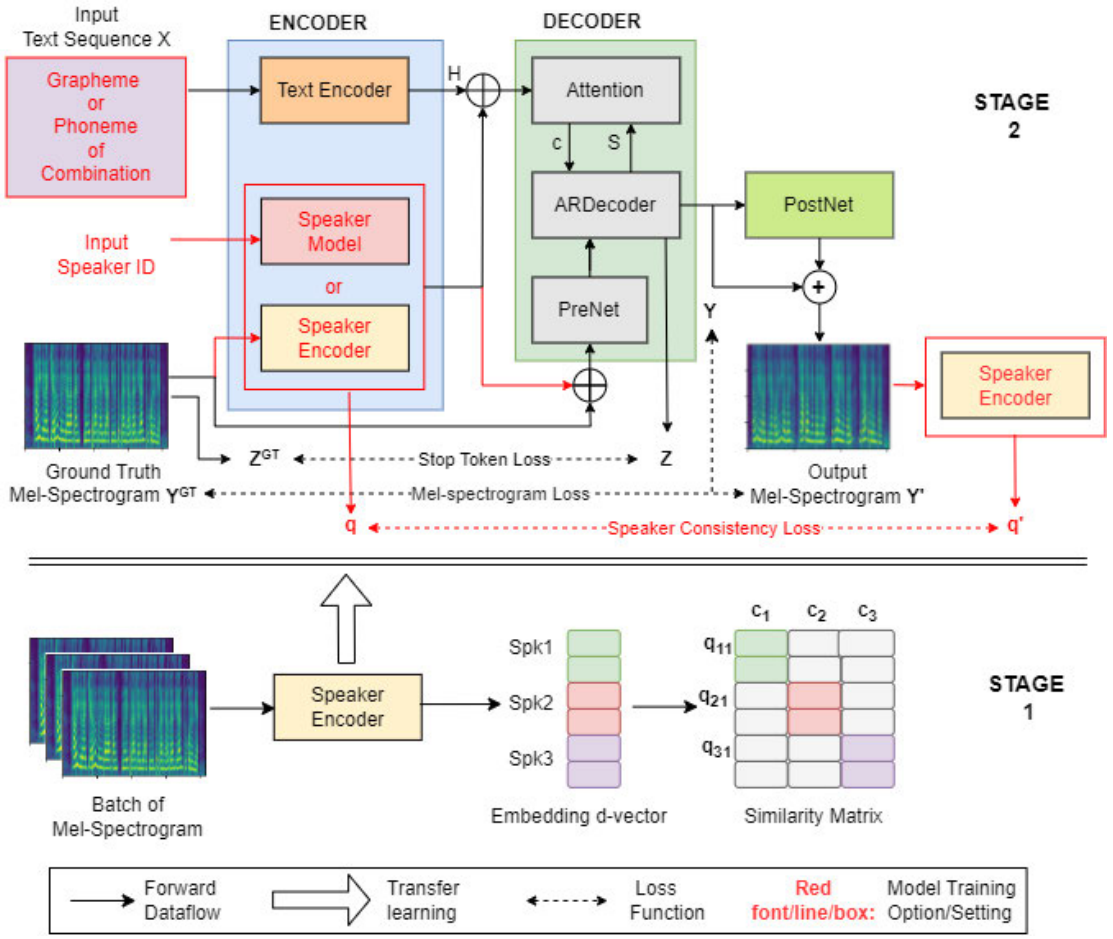


FIGURE 2. Zero-shot voice cloning TTS model training. It consists of two training stages. Stage 1 (bottom) trains the speaker encoder for speaker verification task. Stage 2 (top) transfers the pre-trained speaker encoder weights without fine-tuning to the TTS networks and trains the TTS model for voice cloning speech synthesis task. There are three options for the input-level text sequence: grapheme-level, phoneme-level, or a combination of both with probability 0.5. There are two speaker embedding type options: speaker embedding generated by speaker encoder and speaker embedding using speaker model. There are two alternative positions of the speaker embedding: at encoder only or at both encoder and decoder. There are two loss function options: loss function with or without speaker consistency loss.

The frame-level loss uses three learning objectives: output mel-spectrogram Y and Y' compared to the ground truth mel-spectrogram Y^{GT}, and stop token Z compared to the ground truth stop token Z^{GT}. The model is optimized by minimizing the frame-level reconstruction loss as follows:

$$\mathrm{Loss}_{\mathrm{frame}} = \mathrm{MSE}\left(Y^{\mathrm{GT}}, Y\right) + \mathrm{MSE}\left(Y^{\mathrm{GT}}, Y'\right) + \mathrm{BCE}\left(Z^{\mathrm{GT}}, Z\right) \tag{13}$$

where the summed mean squared error (MSE) and binary cross entropy (BCE) loss functions are used.

The second loss option is to integrate the frame-level reconstruction loss and utterance-level speaker consistency loss as formulated below:

$$\mathrm{Loss}_{\mathrm{total}} = \mathrm{Loss}_{\mathrm{frame}} + \mathrm{Loss}_{\mathrm{speaker}}, \tag{14}$$

where $\mathrm{Loss}_{\mathrm{total}}$ is the total loss function, $\mathrm{Loss}_{\mathrm{frame}}$ is the frame-level reconstruction loss formulated in Eq. (11), and $\mathrm{Loss}_{\mathrm{speaker}}$ is the utterance-level perceptual loss for speaker reconstruction as formulated below:

$$\mathrm{Loss}_{\mathrm{speaker}} = \mathrm{MSE}\left(\boldsymbol{q}, \boldsymbol{q}'\right), \tag{15}$$

where $\boldsymbol{q}$ and $\boldsymbol{q}'$ are the d-vector speaker representations of target speaker and predicted mel-spectrogam after post-net Y', respectively. Unlike model inference that only has one option to generate zero-shot speaker embedding using speaker encoder, the model training process has two options to produce the target speaker embedding representation $\boldsymbol{q}$. The first option is using the speaker encoder network to produce the speaker embedding $\boldsymbol{q}$ from the mel-spectrogram Y^{GT} of the target speaker sample utterance. The speaker representation $\boldsymbol{d}_u$ is used to calculate the speaker embedding $\boldsymbol{q}$ from the average d-vector of all mel-spectrogram windows which can be formulated as follows:

$$\boldsymbol{q} = \boldsymbol{d}_u = \mathrm{SpeakerEncoder}_{\theta_{\mathrm{se}}}(M), \tag{16}$$

where θ_{se} is the model parameters of the speaker encoder function and M is the overall speech mel-spectrogram. The second option uses speaker model $\boldsymbol{d}_{spk}$ for each target speaker based on a number of each speaker's audio recordings as the speaker embedding $\boldsymbol{q}$. We use the speaker model by calculating the average d-vector of each speaker's speech audios as follows:

$$\boldsymbol{q} = \boldsymbol{d}_{spk} == \frac{1}{N_{utt}} \sum_{u=1}^{N_{utt}} \text{SpeakerEncoder}_{\theta_{se}}(M_u), \quad (17)$$

where $\boldsymbol{d}_{spk}$ is the d-vector speaker model representation for speaker spk, N_{utt} is the number of audio recordings, and M_u is the mel-spectrogram of speech utterance u.

Using this training strategy, we can perform zero-shot voice cloning on our TTS model to generate synthesized speech by providing the speaker embedding $\boldsymbol{q}$ generated by the speaker encoder based on the target voice sample.

3) ZERO-SHOT VOICE CLONING TTS MODELS

We propose four categories to develop our models: 1) three options of the input-level of the text sequence, 2) two options of the speaker embedding type, 3) two options of the speaker embedding position, and 4) two options of the loss function. Therefore, based on the number of the options of each category there are 24 models (3×2×2×2=24) in our research. The names of the model for each setting are defined using A-S-P-L- format as follows:

A0S0P0L0	A1S0P0L0	A2S0P0L0
A0S1P0L0	A1S1P0L0	A2S1P0L0
A0S0P1L0	A1S0P1L0	A2S0P1L0
A0S1P1L0	A1S1P1L0	A2S1P1L0
A0S0P0L1	A1S0P0L1	A2S0P0L1
A0S1P0L1	A1S1P0L1	A2S1P0L1
A0S0P1L1	A1S0P1L1	A2S0P1L1
A0S1P1L1	A1S1P1L1	A2S1P1L1

where A- is the input-level of the text (A0:grapheme, A1:phoneme, or A2:combination of grapheme and phoneme with probability 0.5), S- is the speaker embedding type (S0:speaker encoder or S1:speaker model), P- is the speaker embedding position (P0:at TTS encoder only or P1:at both TTS encoder and TTS decoder), and L- is the loss function type (L0:frame-level loss only or L1:frame-level loss combined with utterance-level speaker consistency loss).

Model A0S0P0L0 is used as the baseline model. This model is a Tacotron-2-based zero-shot multi-speaker TTS as in [27] combined by GE2E speaker encoder [32]. Model A0S0P0L0 uses a pre-trained d-vector speaker encoder trained for speaker verification task to generate the speaker representation. An explicit speaker label is not required during the model training because it uses the speaker representation extracted directly from the target speaker mel-spectrogram. Model A0S0P0L0 is able to synthesize speech by cloning the voice of the target speakers not included in the training data.

The differences between model baseline A0S0P0L0 and the rest of our models are in the category setting options in both architecture and the training loss as explained above.

IV. EXPERIMENTS

In this section we present the experiment details including the datasets used, the network implementation, and the metrics used to evaluate our TTS models.

A. DATASETS

To perform this research, we use two datasets as the source domain and the target domain for the zero-shot learning. The source domain dataset is used to train the speaker encoder for speaker verification task and the TTS model for speech synthesis task. Meanwhile, the target domain dataset is used to test the model performance on both normal human and people with dysphonia.

As the source domain to train the speaker encoder and TTS model we use LibriTTS-Train360 dataset [33]. This publicly available dataset is an English speech corpus that consists of 115,596 multi-speaker speech utterances recorded with a sampling rate of 24000 Hz. There are 904 speakers (474 males and 430 females) with a total duration of 357.3 hours.

For the zero-shot voice cloning target domain we use UncommonVoice dataset [2], an English speech corpus that consists of speech utterances from both normal speakers and dysphonia speakers. UncommonVoice dataset is available from the original authors upon request [45]. We use all read speech recordings T2 that consists of 671 speech utterances from 9 normal speakers (3 males and 6 females) and 2521 speech utterance from 35 dysphonia speakers (7 males and 28 females). The speech utterances are recorded with a sampling rate of 16000 Hz with a total duration of 0.8 hours and 4.6 hours for normal speakers and dysphonia speakers, respectively.

Since we use speech sampling rate of 22050 Hz in our voice cloning TTS model, we apply an audio pre-processing to down-sample LibriTTS recordings from 24000 Hz into 22050 Hz and to up-sample UncommonVoice recordings from 16000 Hz into 22050 Hz. The spectral analysis used for each speech utterance is 80 channel mel-spectrogram, Hann window size of 1024, hop size of 256, and 1024-point FFT.

B. NETWORK IMPLEMENTATION DETAILS

We develop the proposed TTS models using open source Tacotron-2 [46]. The models are implemented in Python using the PyTorch library. We make some modifications and network additions including: 1) adding options of the input-level of the text, 2) adding a speaker encoder network for zero-shot voice cloning, 3) adding options of the speaker embedding types, 4) adding options to where the speaker embedding is fed, and 5) adding options of the loss functions used. We also adjust the model hyper-parameters as presented in Table 1. For the speaker encoder network, we use open source speaker encoder code [47] and add

TABLE 1. The model hyper-parameters and spectral analysis of zero-shot voice cloning TTS model.

Spectral analysis	sample rate=22050 Hz; window size=1024;
	window type= Hann; hop size= 256;
	N_FFT=1024; mel channel=80

Character Embedding	1 layer FCNN: FC-512-ReLU
Text Encoder	3 layer CNN: 512 filters with shape 5x1; 1 layer bi-LSTM: 512 units
Language Encoder	1 layer FCNN: FC-8-ReLU
Style Encoder	6 layer CNN: 32, 32, 64, 64, 128, 128 filters with shape 3x3 and stride 2x2; 1 layer GRU: 128 units Multi-head Attention: dim = 256; head= 8; token=10
Speaker Encoder	3 layer LSTM: 384 units 1 layer FCNN: FC-128
Attention	1 layer LSTM: 1024 units Q,K,V layer FCNN: FC-128-tanh 1 CNN: 32 filters with shape 31x1 1 layer FCNN: FC-128-tanh
PreNet	2 layer FCNN: FC-256-ReLU (Dropout=0.1)
ARDecoder	2 layer LSTM: 1024 units
PostNet	5 layer CNN: 512 filters with shape 5x1

the network into our TTS model codes. We make necessary changes on the speaker encoder networks especially the model hyper-parameters and the spectral analyses according to our proposed TTS model. To develop the vocoder we use open source NVIDIA WaveGlow [48] by adjusting some hyper-parameters and the spectral analysis.

We train our models on a single NVIDIA DGX-1 GPU using batch size 64, ADAM optimization [49] with default parameters, learning rate from 1e-3, and weight decay 1e-6.

C. MODEL EVALUATION

To evaluate the speaker similarity, intelligibility, and naturalness of the speech generated by the TTS models, we use three assessments as follows: 1) speaker cosine similarity (COS), 2) character error rate (CER), and 3) mean opinion score (MOS) for speech quality.

Speaker space cosine similarity (COS) is to measure the similarity of the speaker characteristic between the ground truth target speaker and synthesized speech. COS as used in [28] and [36] is calculated as follows:

$$\text{Cos}\left(\boldsymbol{d}_u, \boldsymbol{d}_{\text{spk}}\right) = \frac{\boldsymbol{d}_u.\boldsymbol{d}_{\text{spk}}}{\|\boldsymbol{d}_u\| \left\|\boldsymbol{d}_{\text{spk}}\right\|} = \frac{\sum_{i=1}^{N} \boldsymbol{d}_{u,i}.\boldsymbol{d}_{\text{spk},i}}{\sqrt{\sum_{i=1}^{N} \boldsymbol{d}_{u,i}^2}\sqrt{\sum_{i=1}^{N} \boldsymbol{d}_{\text{spk},i}^2}} \tag{18}$$

where $\boldsymbol{d}_u$ is d-vector speaker representation of the synthetic speech utterance, $\boldsymbol{d}_{spk}$ is the target speaker model, and N are the dimension of the vector representations.

Character error rate (CER) [50] is used to evaluate the intelligibility and clarity of the speech in character-level. We use Whisper [51] to generate text from the speech and compare the output text with the reference text using CER computed using the formula as follows:

$$CER = \frac{S_C + D_C + I_C}{N_C} \tag{19}$$

where S_C is the number of character substitutions, D_C is the number of character deletions, I_C is the number of character insertions, and N_C is the number of whole characters.

Mean opinion score (MOS) is to measure the quality of speech synthesis [52]. MOS evaluation is usually performed by 15-30 respondents to score the presented speech using an absolute scale of 1-5 with 0.5 point increase, 1: bad, 2: poor, 3: fair, 4: good, 5: excellent. However, in this research we use MosNet [53], a deep learning-based objective assessment for voice conversion, to do the evaluation. We use MosNet application instead of real human accessors because it is more practical since we have many speech utterances to be evaluated: (1+24) x (671+2521) speech utterances.

V. RESULTS AND ANALYSES

This section presents the overall performance of the TTS models, the ablation study for each modification category, and the comparison of the human voice, the baseline model, and the best model measured using COS, CER, and MOS evaluation metrics.

A. ZERO-SHOT VOICE CLONING TTS MODELS PERFORMANCE

To evaluate the performance of zero-shot voice cloning TTS models, we use three metrics: COS to measure the speaker cosine similarity between the synthesized speech and the target human voice, CER to measure intelligibility, and MOS to measure speech quality. Table 2 contains the measurement results of the three evaluation metrics for 24 TTS models that are experimented. Model A0S0P0L0 is used as the baseline model. The table also presents an evaluation of the ground truth (GT), the speech voice produced by normal humans and the people with dysphonia. The report is divided into normal human voices and dysphonic human voices. Table 2 also presents the rank of COS, CER, and MOS evaluations for each model compared to others. Then, to find the best model the average rank of all metrics is calculated for each model and a final rank is given based on the average rank.

The cosine speaker similarity evaluation presented in Table 2 shows that all models provide sufficient COS score results. Model A2S1P0L1 provides the best COS evaluation for both normal human voices and dysphonic voices. The model increases the COS score compared to baseline model A0S0P0L0 by 0.155 and 0.197 for normal human voices and people with dysphonic voices.

CER evaluation shows that all models can significantly improve the intelligibility of the dysphonia-to-normal voice cloning speech compared to the original dysphonic voices. However, for the normal voice cloning the CER scores of the synthesized speech has slightly worse compared to

TABLE 2. TTS models evaluation.

Category			Model	Normal			Dysphonia			Normal Rank			Dysphonia Rank			REKAP	
				COS	CER	MOS	COS	CER	MOS	COS	CER	MOS	COS	CER	MOS	AVG	RANK
			GT	0.835	2.69	3.111	0.883	18.13	2.998	-	-	-	-	-	-	-	-
L0: without speaker loss	A0	P0 S0	A0S0P0L0	0.432	6.59	2.826	0.333	5.23	2.974	16	16	20	19	7	21	16.5	19
		P0 S1	A0S1P0L0	0.476	**4.70**	2.985	0.474	**4.28**	3.219	13	**4**	12	7	**2**	8	7.7	4
		P1 S0	A0S0P1L0	**0.576**	5.55	**3.579**	**0.475**	5.57	**3.693**	**3**	7	**1**	**6**	9	**1**	4.5	2
		P1 S1	A0S1P1L0	0.419	7.90	3.033	0.309	4.89	3.187	18	22	9	20	5	10	14	17
	A1	P0 S0	A1S0P0L0	0.408	**4.34**	**2.944**	0.455	5.44	**3.123**	21	**3**	**15**	10	8	**14**	11.8	12
		P0 S1	A1S1P0L0	**0.523**	5.14	2.913	**0.469**	**4.21**	3.114	**7**	5	18	**8**	**1**	15	9	6
		P1 S0	A1S0P1L0	0.412	6.55	2.903	0.307	6.04	2.914	20	15	19	21	13	22	18.3	21
		P1 S1	A1S1P1L0	0.394	10.23	2.799	0.279	7.75	2.990	23	24	22	23	22	20	22.3	23
	A2	P0 S0	A2S0P0L0	0.386	6.50	2.767	0.368	5.95	2.872	24	13	23	18	12	23	18.8	22
		P0 S1	A2S1P0L0	0.415	**5.47**	**3.318**	**0.434**	**5.21**	**3.388**	19	**6**	**2**	**12**	**6**	**4**	8.2	5
		P1 S0	A2S0P1L0	**0.452**	6.35	3.259	0.303	6.96	3.333	**15**	12	5	22	17	5	12.7	16
		P1 S1	A2S1P1L0	0.396	10.00	2.679	0.277	7.13	2.818	22	23	24	24	19	24	22.7	24
L1: with speaker loss	A0	P0 S0	A0S0P0L1	0.503	6.73	**3.308**	0.439	8.69	3.453	10	17	**3**	11	24	3	11.3	11
		P0 S1	A0S1P0L1	0.523	7.51	3.102	**0.496**	8.28	3.256	7	20	6	**3**	23	6	10.8	9
		P1 S0	A0S0P1L1	**0.580**	**6.24**	3.294	0.478	**6.73**	**3.506**	**2**	**11**	4	5	**15**	**2**	6.5	3
		P1 S1	A0S1P1L1	0.426	6.86	2.953	0.419	7.20	3.085	17	19	14	14	20	17	16.8	20
	A1	P0 S0	A1S0P0L1	0.477	**4.24**	2.802	0.459	5.58	3.000	12	**2**	21	9	10	19	12.2	13
		P0 S1	A1S1P0L1	**0.534**	6.12	2.919	**0.485**	5.73	3.103	**4**	9	17	**4**	11	16	10.2	7
		P1 S0	A1S0P1L1	0.495	5.86	2.959	0.395	**4.78**	3.136	11	8	13	16	**4**	13	10.8	10
		P1 S1	A1S1P1L1	0.468	7.56	**3.015**	0.392	7.03	**3.139**	14	21	**10**	17	18	**12**	15.3	18
	A2	P0 S0	A2S0P0L1	0.523	6.85	2.937	0.528	6.25	3.049	7	18	16	2	14	18	12.5	15
		P0 S1	A2S1P0L1	**0.587**	**3.94**	3.010	**0.530**	**4.68**	3.218	**1**	**1**	11	**1**	**3**	9	**4.3**	**1**
		P1 S0	A2S0P1L1	0.524	6.20	**3.064**	0.421	7.43	**3.224**	5	10	**7**	13	21	**7**	10.5	8
		P1 S1	A2S1P1L1	0.521	6.55	3.060	0.416	6.80	3.178	9	14	8	15	16	11	12.2	14

Note: GT is ground truth speech from both normal human speech and dysphonic human speech. Evaluation: COS is the cosine similarity (higher score is better), CER is the character error rate in % (lower score is better), MOS is mean opinion score for speech quality (higher score is better), Evaluation Rank is a model evaluation score rank compared to other models' scores (lower rank is better), AVG is the average rank of a model evaluation scores, RANK is the rank of the average rank. Category: loss function (L0: no speaker loss, L1: with speaker loss), input level of the text (A0: grapheme, A1: phoneme, A2: combination), speaker embedding position (P0: at TTS encoder, P1: at TTS encoder and decoder), speaker embedding type (S0: speaker embedding, S: speaker model).

that of the original human voice GT. Similar to the COS evaluation, model A2S1P0L1 also provides the best CER score for normal voice cloning TTS with an error decrease of 2.65% compared to the baseline model A0S0P0L0 and a slight error increase of 0.98% compared to the normal human voice GT. Different results are found in voice cloning TTS for human with dysphonia. In this case, CER score of model A2S1P0L1 (4.28%) is slightly worse than of model A1S1P0L0 (4.21%) which gives the best CER score. Model A1S1P0L0 can improve CER score by 1.02% and 13.92% compared to the baseline model A0S0P0L0 and the original voice of human with dysphonia, respectively.

The MOS evaluation for speech quality in Table 2 shows that the majority of the TTS models give better scores than the original voices of humans with dysphonia, but slightly worse than the original voices of normal humans. Model A0S0P1L0 produces synthesized speeches with the best MOS score for both normal humans and dysphonia sufferers voice cloning. In normal human voice cloning TTS, model A0S0P1L0 increases the MOS scores by 0.753 and 0.468 compared to the baseline model A0S0P0L0 and the original human speech

GT, respectively. In the dysphonia-to-normal voice cloning TTS, this model increases MOS by 0.719 and 0.695 compared to the baseline model and the original dysphonic voices GT.

The overall performance of the models is calculated using the rank (RANK column) of the score ranks average (AVG column) presented in Table 2. The best model is model A2S1P0L1 with a score ranks average value of 4.3, followed by model A0S0P1L0 and then model A0S0P1L1. The best model A2S1P0L1 is a zero-shot TTS that applies the following setting in each category: uses input-level in the form of a combination of grapheme and phoneme (A2), generates speaker embeddings using speaker model (S1), puts the speaker embeddings only at the TTS encoder (P0), and applies a combined loss function of frame-level loss and utterance-level speaker consistency loss (L1).

Furthermore, to find out the contribution of each category setting to the performance of zero-shot voice cloning TTS models, we present an ablation study for each category in the following four sections. As previously explained we propose four categories of settings: input-level of the text sequence, speaker embedding type, speaker embedding position and loss function.

B. ABLATION STUDY ON THE INPUT-LEVEL OF TEXT SEQUENCE

To know the contribution of the input-level text sequence category to the model performance, we compared the evaluation of grapheme-level (A0), phoneme-level (A1), and grapheme-phoneme combination (A2) head-to-head by grouping 24 models into groups with the same settings for other categories. Figure 3 presents bar graphs that compare the input-level options on three evaluations: COS (top), CER (middle) and MOS (bottom).

From the COS bar graphs in Figure 3, it cannot be concluded which input-level option gives the best performance without looking at the cases and settings from other categories. In the case of normal human voice cloning, grapheme-level (A0) gives the best speaker cosine similarity in average AVG-N, while in the case of people with dysphonia grapheme-level (A0) and combination (A2) have quite the same performance. If we look in more detail on the model grouping by loss function, we can see that TTS voice cloning for both normal human and dysphonia cases shows consistent result depending on the loss function settings. The graph shows that grapheme-level (A0) gives the best performance when we apply the loss function without speaker consistency loss (L0), whereas when we apply speaker consistency loss (L1) the best speaker cosine similarity is given by the grapheme-phoneme combination (A2). This can be seen from the bar graph of each model group as well as the average value of the model group AVG-L0 and AVG-L1 for normal human and dysphonia cases, respectively.

The CER bar graph also shows similar results. In average, grapheme-level (A0) has the best performance for normal human case, while phoneme-level (A1) and combination (A2) provide the best performance for dysphonia case. Likewise in relation to the loss function setting, for model without speaker loss (L0) the best performance is achieved by grapheme-level (A0) for both normal human and dysphonia cases, while for model with speaker loss (L1) the best performance is achieved by combination-level (A2) for normal human case and phoneme-level (A1) for dysphonia case.

The MOS bar graph evaluation has slightly different results from COS and CER as shown in Figure 3. In general, the grapheme-level input text sequence (A0) provides the best MOS score in most model groups. However, there is an exception found in the A-S1P0L0 model group where combination-level (A2) provides better performance in normal human case for both with and without speaker consistency loss.

C. ABLATION STUDY ON SPEAKER EMBEDDING TYPE

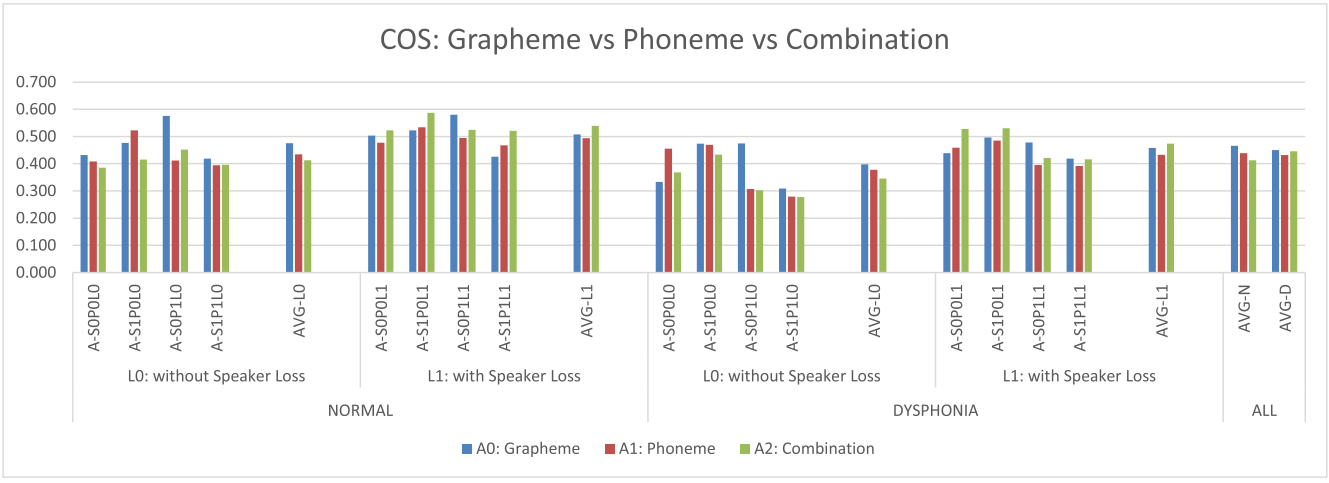
As explained previously, we apply two types of speaker embedding during training: speaker embedding generated by the speaker encoder (S0) and by finding a speaker model (S1) for each speaker. To find out which speaker embedding type is better, we also group 24 models in the way that models with the same settings for the other categories are in the same group so that we can compare the performance of the speaker encoder (S0) and speaker model (S1) directly, as presented in Figure 4.

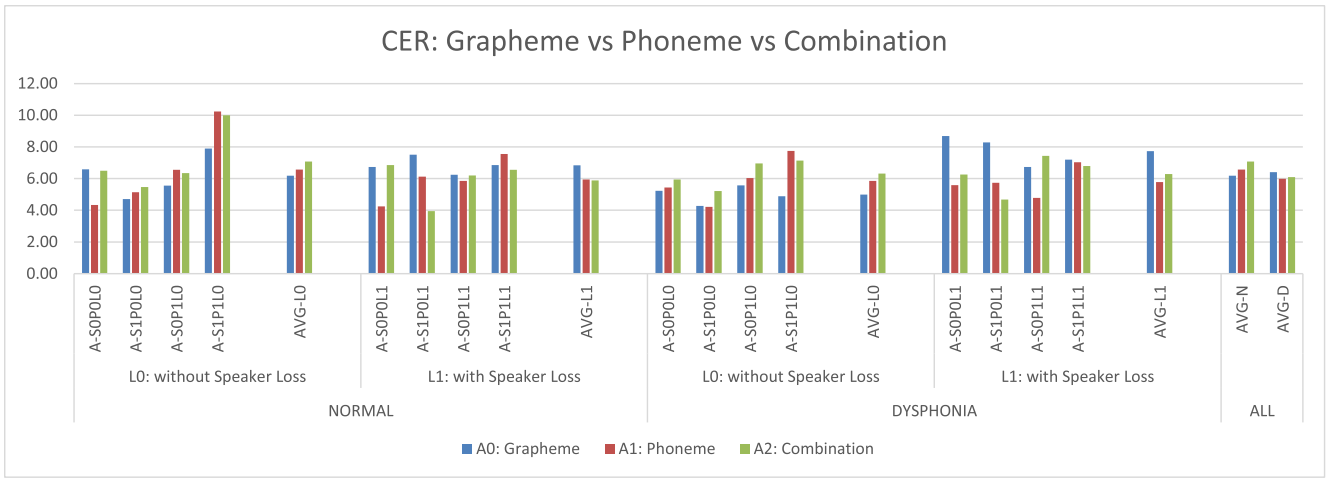
Interesting results are provided by the comparison of COS bar graphs based on the speaker embedding types. The average of models' speaker cosine similarity for both normal human AVG-N and dysphonia AVG-D shows that the speaker encoder (S0) and speaker model (S1) have quite the same performance. Interestingly, if we look further it seems that there is a connection between the speaker embedding types and the speaker embedding positions. In the model group with speaker embedding position at the TTS encoder (P0), a better speaker similarity result is achieved by the speaker model (S1). Meanwhile, in the model group with speaker embedding position at the TTS encoder and decoder (P1), using the speaker encoder (S0) provides better performance than that of the speaker model (S1). This can be seen from all model bar graphs including the model group averages AVG-P0 and AVG-P1 for both normal human and dysphonia voice cloning cases.

Similar results are also shown in the CER and MOS bar graphs, where the speaker model (S1) gives better results if the speaker embedding position is only at the TTS encoder (P0) than that of the speaker encoder (S0). However, if the position of the speaker embedding is at the TTS encoder and decoder (P1), then the speaker encoder (S0) performs better than the speaker model (S1).

D. ABLATION STUDY ON SPEAKER EMBEDDING POSITION

This section explains the effect of speaker embedding position. To compare the performance of the speaker embedding





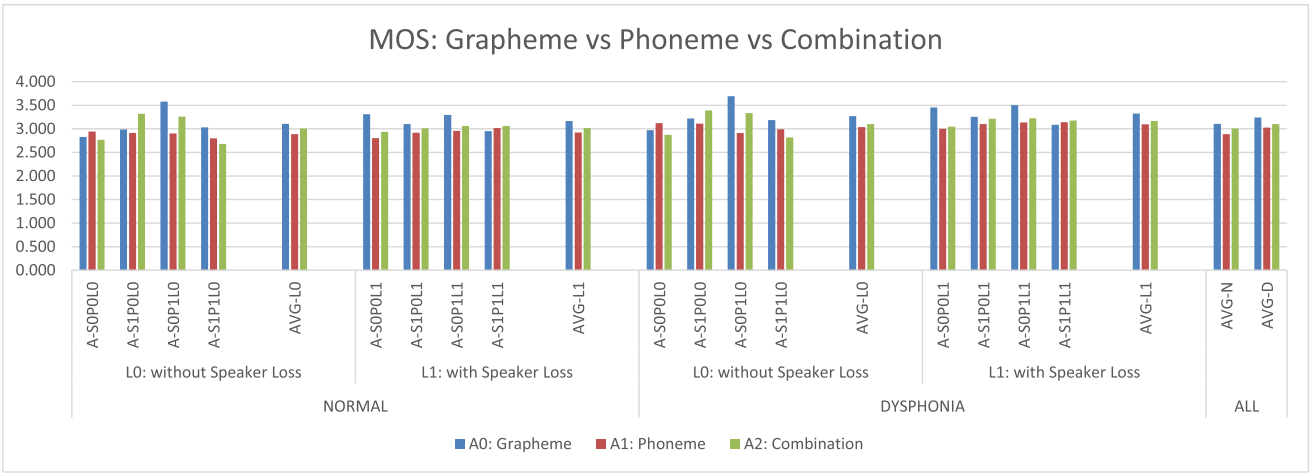


FIGURE 3. COS, CER, and MOS evaluation comparison based on input-level of the text sequence, A0: grapheme, A1: phoneme, A2: combination of grapheme and phoneme. COS bar graph (top): the higher score the better. CER bar graph (middle): the lower score the better. MOS bar graph (bottom): the higher score the better.

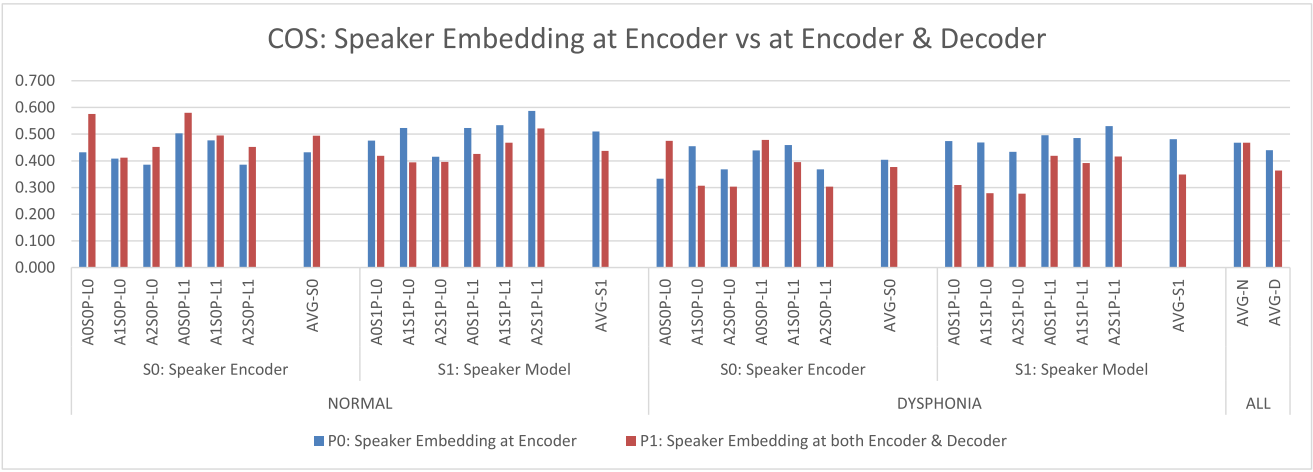
position at both TTS encoder (P0) and TTS encoder and decoder (P1), we arrange the 24 models into model groups with the same settings for the other categories. Figure 5 presents bar graphs for COS (top), CER (middle) and MOS (bottom) evaluations based on the speaker embedding positions.

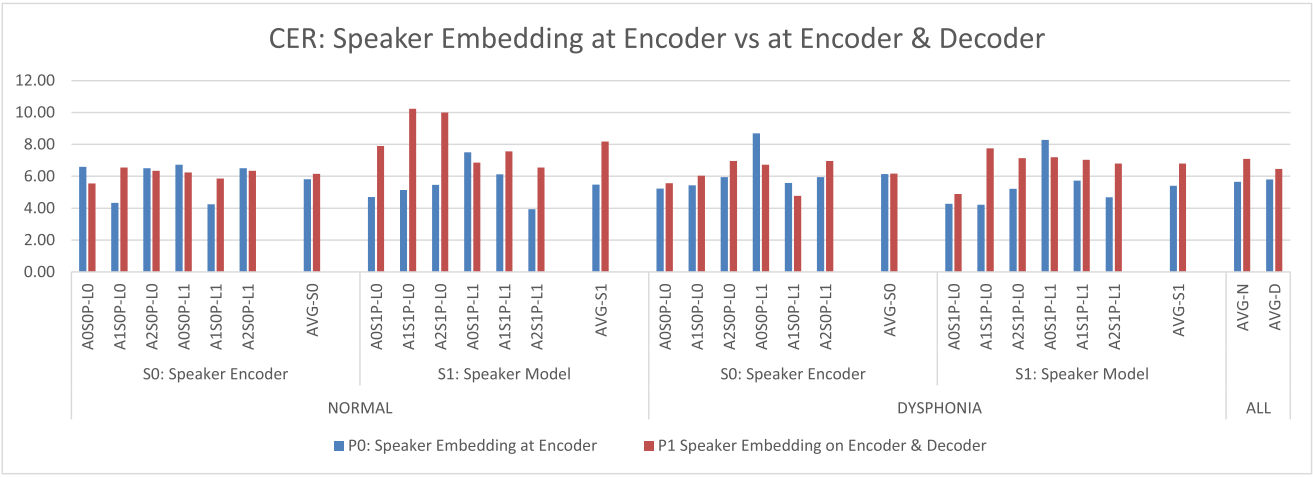
The COS bar graph shows that both speaker embedding positions provides quite the same performance for normal



FIGURE 4. COS, CER, and MOS evaluation comparison based on speaker embedding type, S0: speaker encoder, S1: speake model. COS bar graph (top): the higher score the better. CER bar graph (middle): the lower score the better. MOS bar graph (bottom): the higher score the better.

human voice cloning AVG-N. This is not the case for dysphonia voice cloning as shown by the bar height of the model average AVG-D. For dysphonia case, the speaker embedding position at the TTS encoder (P0) give better speaker cosine similarity than that of the TTS encoder and decoder (P1). In accordance with the results of the ablation study in Section V-C which shows that there is a relationship between the type and the position of the speaker





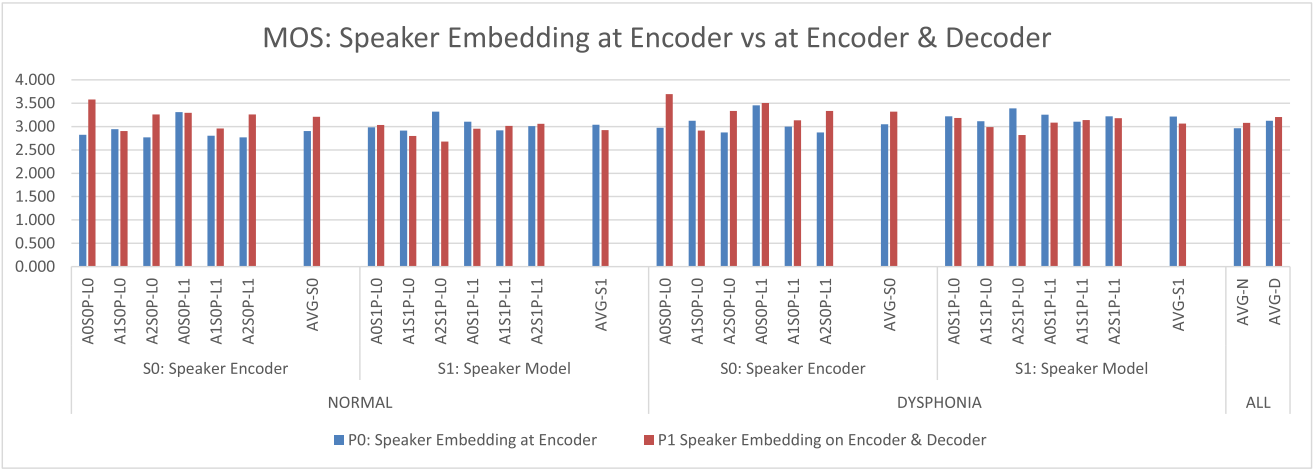


FIGURE 5. COS, CER, and MOS evaluation comparison based on speaker embedding position, P0: at TTS encoder, P1: at TTS encoder and decoder. COS bar graph (top): the higher score the better. CER bar graph (middle): the lower score the better. MOS bar graph (bottom): the higher score the better.

embedding, we divide the models into two speaker embedding type groups and analyze each model group. For normal human case, a better performance is achieved by the TTS models with speaker embedding positions at the encoder and decoder (P1) for model group with the encoder speaker (S0), whereas for the model group with speaker model (S1) the speaker embedding position at the TTS encoder (P0) performs better. As for dysphonia voice cloning case in both model group with the speaker encoder (S0) and speaker model (S1), the speaker embedding position at the TTS

encoder (P0) provides better result than that of the TTS encoder and decoder (P1). This can be seen from the value of AVG-S0 and AVG-S1 for both normal human and dysphonia voice cloning cases.

As shown in Figure 5, the CER evaluation result is also in line with the CER evaluation result. For the normal human voice cloning case, the speaker embedding position on the TTS encoder and decoder (P1) produces better performance in the model group with the encoder speaker (S0), while the position on the TTS encoder (P0) performs better on the model group with speaker model (S1). For the dysphonia voice cloning, the speaker embedding position in the TTS encoder (P0) provides better character error rate than that of the TTS encoder and decoder (P1) in both group models with the speaker encoder (S0) and speaker model (S1).

The MOS bar graph in Figure 5 shows consistent results for both normal human and dysphonia voice cloning cases. The speaker embedding at TTS encoder and decoder (P1) performs the best in the model group with the speaker encoder group (S0), while in the model group with speaker model (S1) the best speech quality is achieved by the speaker embedding position at the TTS encoder only (P0). However, if we look at the overall models average for both normal human AVG-N and dysphonia AVG-D, the speaker embedding position at the TTS encoder and decoder (P1) is slightly better than the position at the TTS encoder only (P0).

E. ABLATION STUDY ON LOSS FUNCTION

The final ablation study is to determine the effect of the loss functions used: loss function without speaker loss (L0) and loss function with speaker loss (L1). As in the ablation study of other categories explained in the previous sections, we also present 24 models into several groups with the same settings for other categories in order to compare the performance of both loss functions head-to-head. Figure 6 presents bar graphs of COS (top), CER (middle) and MOS (bottom) evaluations based on the loss functions.

From the COS bar graph, it can be seen that the model using speaker loss (L1) consistently provides a better speaker similarity than that of without speaker loss (L0). This happen in all model groups for both normal human and dysphonia voice cloning cases. The MOS evaluation bar graph also shows similar results to the COS evaluation, where model with speaker loss (L1) produces better speech sound quality compared to that of without speaker loss (L0).

However, different results can be seen in the CER evaluation. The use of speaker loss (L1) does not always provide a better speech recognition result than that of without speaker loss (L0). The CER bar graph in Figure 6 shows that the speaker loss performance can be related to the input level used. If the model uses grapheme-level text sequence (A0), it turns out that speaker loss (L1) actually reduces the speech intelligibility measured by CER, whereas if the model uses phoneme-level (A1) or combination (A2), the model with speaker loss (L1) has a better CER performance compared to the model without speaker loss (L0).

F. DYSPHONIC SPEAKERS SPEECH, BASELINE MODEL, AND THE BEST MODEL VOICE CLONING COMPARISON

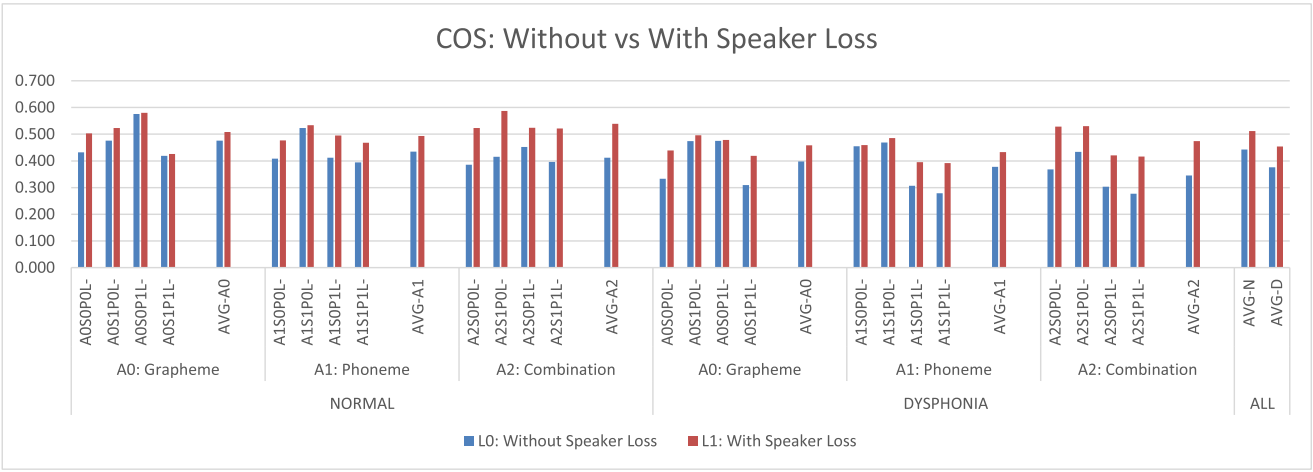
After explaining the overall performance of our TTS models including ablation studies for each setting category in the previous sections, this section shows a comparison of evaluation results from several voice samples produced by original human having dysphonia, the baseline TTS model, and the best proposed voice cloning TTS model. Three evaluation metrics are used: COS for speaker similarity, CER for speech intelligibility, and MOS for speech quality. As explained in Section IV-A, the best model based on the total ranking of the three evaluation metrics is model A2S1P0L1, while the baseline used is model A0S0P0L0.

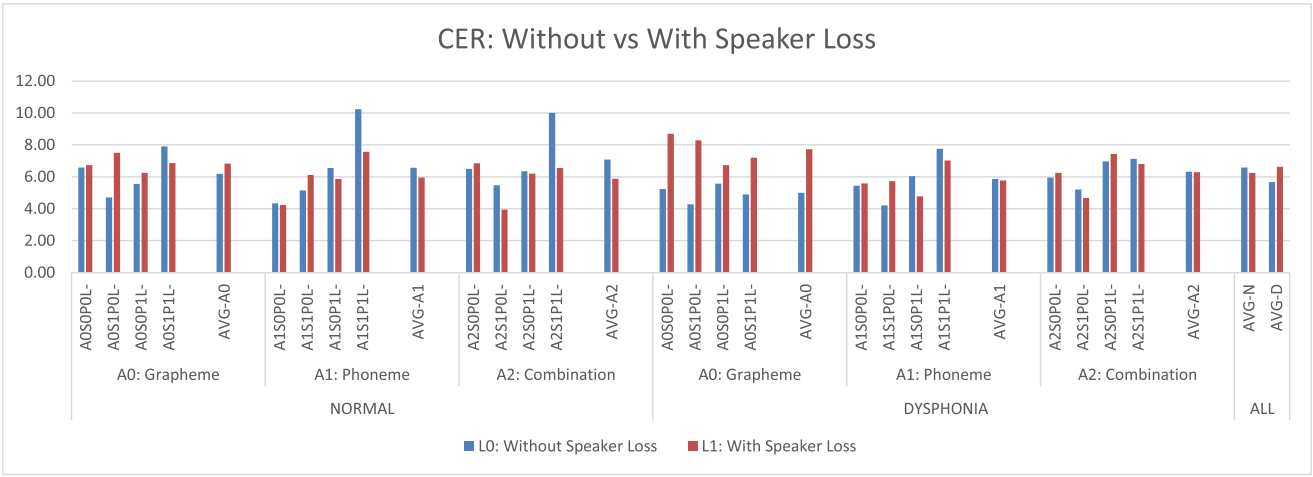
To provide a better picture of the performance comparison from the three speech sources that produce speech samples, we present COS, CER, and MOS evaluation and a mel-spectrogram visualization in Figure 7. The original human speech samples are taken from two female and two male speakers suffering from dysphonia speech disorder. These speech samples are then compared with voice cloned speeches produced by the baseline model and the best model for the same input text.

1) SPEAKER COSINE SIMILARITY

The speaker similarity is measured by calculating the cosine between the d-vector of each speech sample and the d-vector of speaker model as presented in Eq. (18). The speaker model is the centroid of all provided speech utterances for each speaker as shown in Eq. (17). From the quantitative COS evaluation presented in Table 2, the best model A2S1P0L1 is able to increase cosine similarity by 0.197 compared to the baseline model A0S0P0L0. This means that the best model is able to produce cloned voices with a higher similarity to the original speaker's voice compared to the baseline. For example, the four speech samples presented in Figure 7 shows that the COS values from the best model (right) are higher than that of the baseline model (middle).

The visualization of mel-spectrogram in Figure 7 provides rough estimate of the fundamental frequency (F0) contour. F0 is the main acoustic dimension that relates to the speaker identity. Typically F0 lie roughly in the range 80 to 450 Hz, where males have lower voices than females and children. F0 contour tends to follow the lowest frequency with the most energy, which is indicated by bright colors toward the bottom of the mel-spectrogram image. The dysphonic female speech samples (a) and (b) show that F0 of the best model has similar height as F0 of the original human voice, while F0 of the baseline model is lower than that of both. As the result, in many speech samples the baseline model produces voice cloning speech sounds that are not similar to the original human voice. Moreover, some female voice cloning speech samples produces by the baseline model have too low F0 resulting in the cloned speech sound resembles a man's voice.





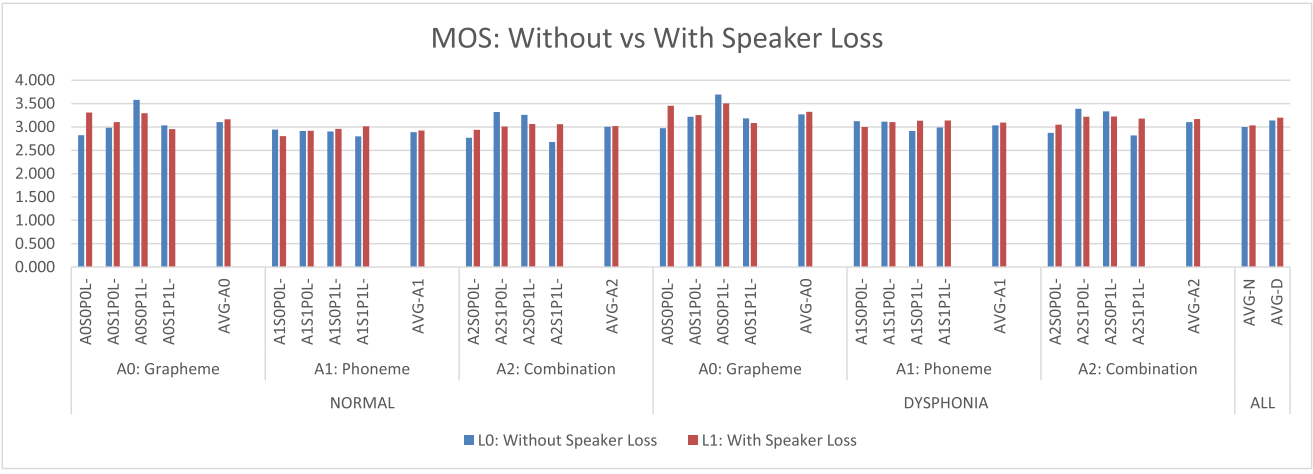
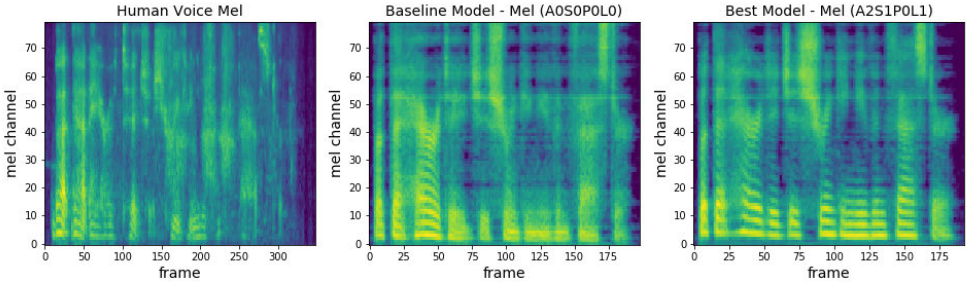


FIGURE 6. COS, CER, and MOS evaluation comparison based on loss function, L0: without speaker loss, S1: with speaker loss. . COS bar graph (top): the higher score the better. CER bar graph (middle): the lower score the better. MOS bar graph (bottom): the higher score the better.

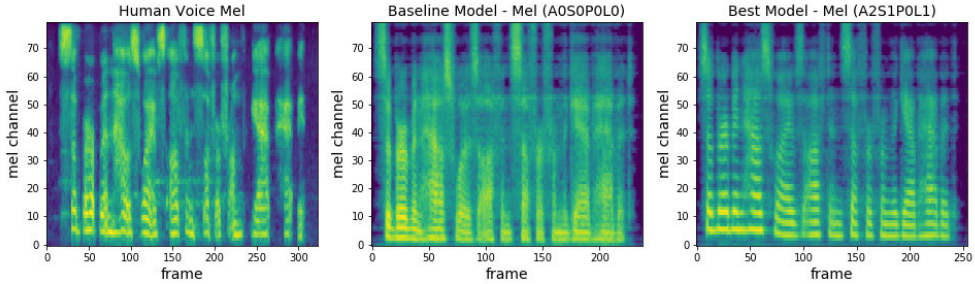
This is not the case for the best proposed TTS model that is able to synthesize a more similar cloned speech to the original human voice.

To give a better illustration of the speaker similarity we provide the visualization of the d-vector speaker space in Figure 8. We use uniform manifold approximation and projection (UMAP) [54] to visualize the d-vector speaker representation in two dimensional vector space. Figure 8 presents speech samples of five female speakers (bold color) and five male speakers (soft color). If we look at the vector



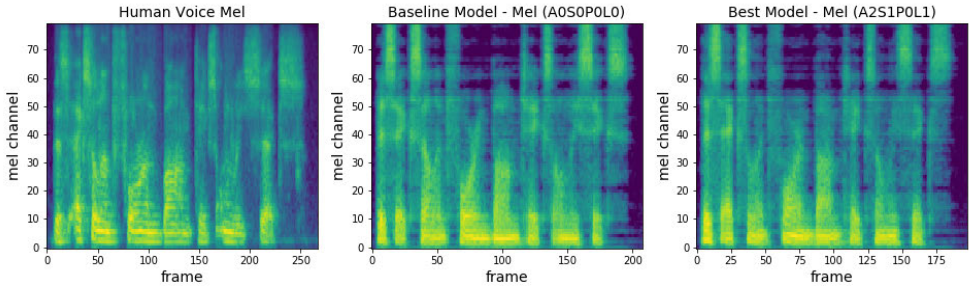
Human Voice Mel	Baseline Model - Mel (A0S0P0L0)	Best Model - Mel (A2S1P0L1)
ASR: but that air of tiches is shrinking faster COS: 0.653 CER: 35.71 MOS: 3.008	ASR: but that heritage is shrinking even faster COS: 0.413 CER: 0 MOS: 3.067	ASR: but that heritage is shrinking even faster COS: 0.716 CER: 0 MOS: 3.673

(a) Female Speaker FD08 on text "but that heritage is shrinking even faster"



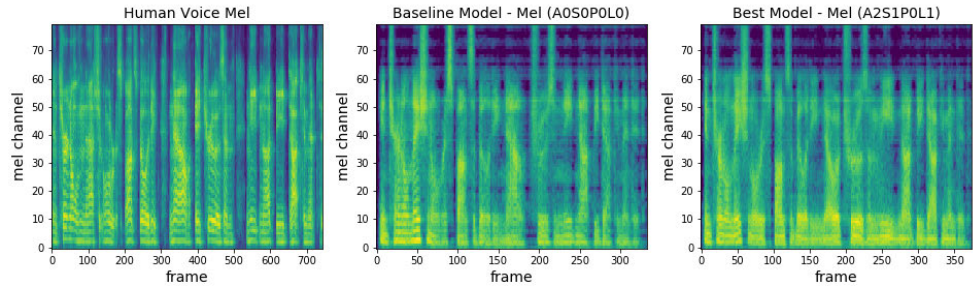
Human Voice Mel	Baseline Model - Mel (A0S0P0L0)	Best Model - Mel (A2S1P0L1)
ASR: suburban housewives often suffer from the app happy COS: 0.943 CER: 11.76 MOS: 2.905	ASR: suburban housewives often suffer from the gag habit COS: 0.253 CER: 1.96 MOS: 2.651	ASR: suburban housewives often suffer from the gab habit COS: 0.594 CER: 0 MOS: 3.05

(b) Female Speaker FD035 on text "suburban housewives often suffer from the gab habit"



Human Voice Mel	Baseline Model - Mel (A0S0P0L0)	Best Model - Mel (A2S1P0L1)
ASR: this excellent foreign bond takes only 6 COS: 0.945 CER: 5 MOS: 2.719	ASR: this excellent foreign bomb takes only 6 COS: 0.446 CER: 0 MOS: 3.071	ASR: this excellent foreign bomb takes only 6 COS: 0.537 CER: 0 MOS: 3.205

(c) Male Speaker MD05 on text "this excellent foreign bomb takes only six"



Human Voice Mel	Baseline Model - Mel (A0S0P0L0)	Best Model - Mel (A2S1P0L1)
ASR: suburban housewives often suffer from the app happy COS: 0.922 CER: 5.88 MOS: 2.767	ASR: internal national responsibility now a true is need not be documented COS: 0.669 CER: 4.41 MOS: 2.543	ASR: internal national responsibility no a truism need not be documented COS: 0.767 CER: 1.47 MOS: 3.053

(d) Male Speaker MD07 on text "internal national responsibility now a truism need not be documented"

FIGURE 7. Mel-spectrogram visualization and evaluation of speech samples produced by two female (a and b) and two male speakers (b and c) suffering from dysphonia disorder (left), the TTS baseline model (middle), and the best proposed voice cloning TTS model (right). The evaluation includes the predicted text using automatic speech recognition (ASR) Whisper, the speech intelligibility using character error rate (CER), the speaker cosine similarity (COS), and the speech quality (MOS) using MOSNet. Higher COS is better, lower CER is better, and higher MOS is better.

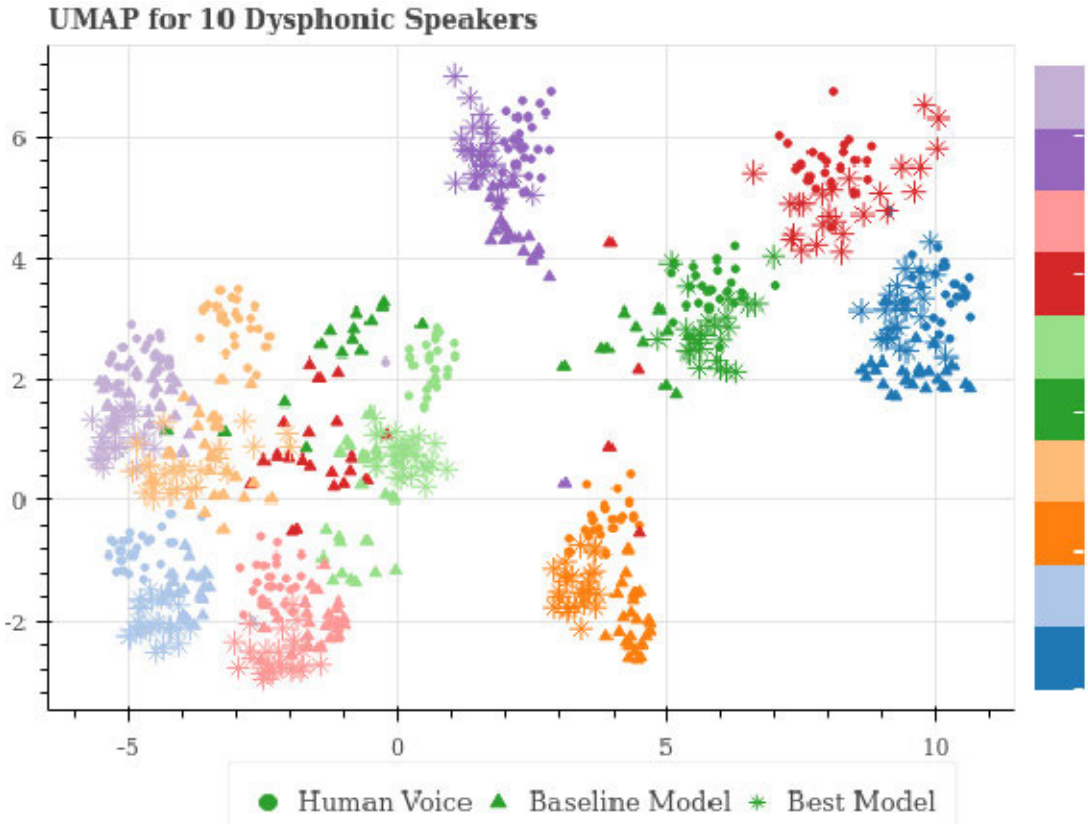


FIGURE 8. The D-vector space visualization of the speaker embedding from speech samples produced by five female and five male speakers suffering from dysphonia speech disorder (circle), voice cloning TTS baseline model (triangle), and the best proposed voice cloning TTS model (asterisk). Different colors indicate different speakers. Bold colors indicate female speakers while soft colors indicate male speakers.

space, male speakers is able to be separated from female speakers. However, for some speakers, the d-vector speaker speech representation produced by the baseline model (triangle) is quite far from the d-vector speaker representation of the original human voice (circle). Moreover, several synthetic female voice samples with bold red and bold green triangles are even close to the d-vector space of male speakers (light colors). This does not happen in our best proposed TTS model that is able to produce voice cloned speech with a d-vector speaker representation (asterisk) that is close to the original human voice (circle).

2) SPEECH INTELLIGIBILITY

The speech intelligibility is measured by comparing the speech utterances recognition using Whisper, a state-of-the-art automatic speech recognition application (ASR). Whisper provides the predicted text for each speech utterance produced by human, the baseline model, and the best voice cloning TTS model. The predicted text is then compared to the original text using character-level comparison formulated in Eq. (19) to calculate character error rate (CER) score. We apply text normalization during CER calculation. The CER evaluation presented in Table 2 shows that the best model is able to improve the speech intelligibility by reducing the CER value as much as 13.45% and 0.55% compared to the original dysphonic speaker voice and the baseline model, respectively.

Figure 7 provides four speech samples of input text spoken by dysphonic speakers (left) and synthesized by both the baseline (middle) and the best proposed voice cloning TTS models (right). In general, the four examples (a, b, c, and d) show that: 1) both TTS models produce speeches of which the predicted text has better CER than the original dysphonic speakers voice, and 2) the best TTS model produces better text predictions than that of the baseline model. For a more detailed analysis, let's look at example (a) which uses the text input "but that heritage is shrinking even faster". In this example, the speech utterance spoken by a female dysphonic speaker is recognized by Whisper as the predicted text "but that air of tiches is shrinking faster" with a CER score of 35.71%. This dysphonic speaker have difficulties in producing speech sounds resulting in unclear speech or even unheard speech word, such as the word "even", because of the speaker's interrupted and heavy breath. This is different from voice cloning speech sounds generated by both baseline and our best TTS model that can be recognized by Whisper without any error. The mel-spectrogram visualization in example (a) also shows that the spectrum formants (F1, F2, F3, F4,...)

produced by both TTS models is more clear than that of the dysphonic speech. The formants, indicated by brighter bands running horizontally across the mel-spectrogram, are the main acoustic dimension of verbal difference that affect the intelligibility and clarity of the speech.

3) SPEECH QUALITY

To measure the speech quality we use MOSNet application that has highly correlated with human MOS ratings without the need for expensive human rating. MOSNet assesses the speech quality including the clarity and naturalness. From the MOS evaluation presented in Table 2, the best model is able to increase speech quality by 0.22 and 0.244 compared to the original dysphonic speaker voice and the baseline model, respectively. The baseline model provides a slightly worse MOS score that the human voice. The results can also be shown from the mel-spectrogram plot presented in Figure 7. The best model has the most clear periodic voiced spectrum and less noise unvoiced spectrum in the mel-spectrogram plot. To give a better comparison of the speech sounds we also present a demo of speech samples that can be accessed online.[1]

VI. CONCLUSION

This research presents a zero-shot voice cloning Tacotron-2 based TTS for people with dysphonia. In this case, a zero-shot learning is a necessity because it is impossible to train a TTS model using training data from the speech of dysphonia sufferers. Thus, we use LibriTTS-Train360 dataset as the source domain to train both the speaker encoder network and the TTS model. Next, the trained model is used to synthesize speech sounds in UncommonVoice dataset as the target domain for both normal human and dysphonia voice cloning. The zero-shot TTS model synthesizes speech sound based on input text by performing voice cloning of the target speaker. In this study, we propose four categories: input-level text sequence, speaker embedding type, speaker embedding position, and loss function. Based on each setting option from each category, this research develop 24 different TTS models, one of which is used as the baseline model.

We also present the ablation study for each proposed category. The experimental results conclude that the input-level text sequence category is related to the loss function category used. Grapheme-level performs better if combined with the loss function without speaker loss, while phoneme-level and combination-level perform well combined with the loss function with speaker loss. It can also be concluded that the speaker loss improves the speaker cosine similarity in all model groups, but it decreases the character recognition rate for model group applying grapheme-level while we can maintain the rate by applying phoneme-level or combination-level. Furthermore, the speaker embedding type category is interrelated with the speakers embedding position category. The setting pairs that provide better results are as follows: 1) using the speaker encoder with speaker embedding position at TTS encoder and decoder, and 2) using speaker model with speaker embedding position at TTS encoder. Of the 24 models built, the best model is achieved by proposed model that applies input-level in the form of grapheme-phoneme combination, uses the speaker model to generate the speaker embedding, places the speaker embedding at the TTS encoder only, and uses a combined loss function of frame-level loss and utterance-level speaker loss.

Furthermore, to provide a better understanding of the model performance we present the evaluation analysis for some speech samples. We compare the speech produced by dysphonic speakers, the baseline model, and the best proposed TTS model using COS, CER, and MOS evaluation. We also presents the visualization of the mel-spectrogram plot and d-vector speaker space. Moreover, we provide the speech samples demo online.

From the COS, CER and MOS evaluation results, it can be seen that there is still room for improvement. In general, the evaluation of cosine speaker similarity COS is still not very satisfactory. In future work, we can look for a better speaker representation than the d-vector speaker embedding used in this research. Meanwhile, for the CER evaluation, its performance is quite good and is able to increase the level of speech clarity compared to the original voice of dysphonia sufferers. To improve speech quality measured using MOS evaluation, we can try other neural TTS architectures, such as VITS [17], GlowTTS [25], and FastPitch [26] that are reported in recent work by [55] to have better MOS scores than that of Tacotron-2, which we use in our work. The work [55] compared 12 latest state-of-the-art TTS models, including Tacotron-2 that was reported to have MOS score of 3.09 on TIMIT dataset for normal speakers. This result is slightly higher than our baseline Tacotron-2 based TTS model on dysphonic speakers with the MOS score of 2.974 and lower than our best TTS model with the MOS score of 3.218. Work [55] reported that VITS, GlowTTS, and FastPitch have MOS score of 3.69, 3.54, and 3.48, respectively. By using these TTS models as the base of our zero-shot voice cloning TTS, it is expected to have a higher MOS on our baseline model and even more improvement on our best proposed model.

[1] https://zs_voicecloning.jarkom.cs.ui.ac.id/

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